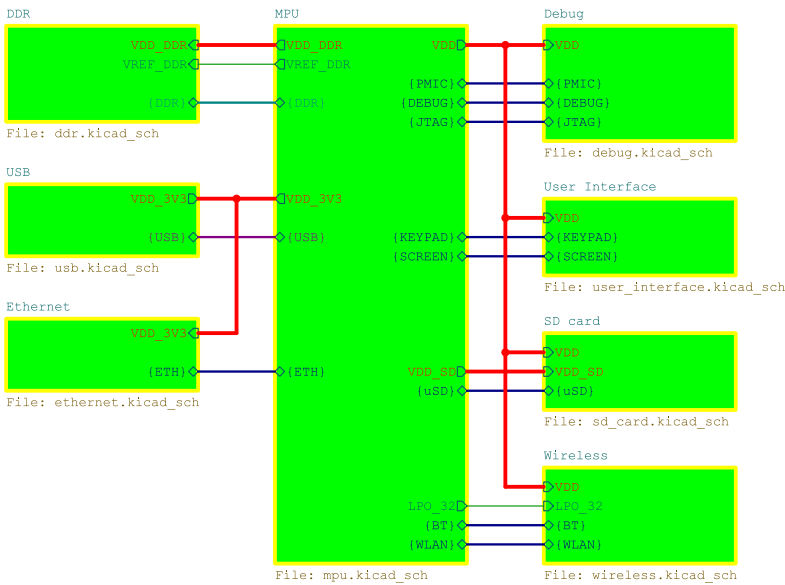
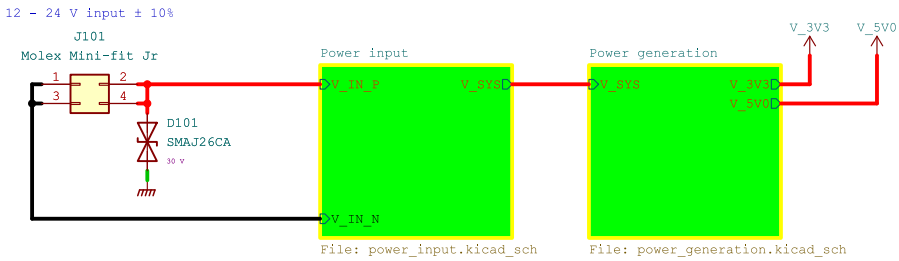


1. Introduction

Table of Contents

- 1. Introduction
- 2. Power input
- 3. Power generation
- 4. MPU
 - 4.1. MPU PMIC
 - 4.2. MPU power
- 5. Debug
- 6. DDR
- 7. USB
- 8. Ethernet
- 9. SDMMC
- 10. User Interface
- 11. Wireless

Block diagram



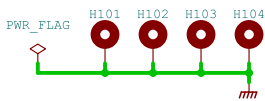
Description

This project serves as a practice exercise in PCB design for a Linux Single-Board-Computer.

Notes

- Schematic comments are in blue
- Layout comments are in magenta
- Mechanical comments are in green

Mechanical



Place fiducials at the top-left, bottom-left, and bottom-right corners of the PCB.

Place a stitched chassis guard rail along the edges of the PCB to provide a safe point for human contact when handling the device.

Sheet: /
File: linux-sbc.kicad_sch

Title: Linux SBC

Size: A3 Date: 2025-11-07

KiCad E.D.A. 9.0.6

Rev: v0.0.1

Id: 1/13

2. Power input

Protection

Reverse polarity insertion protection and a resettable fuse for continous significant overcurrent

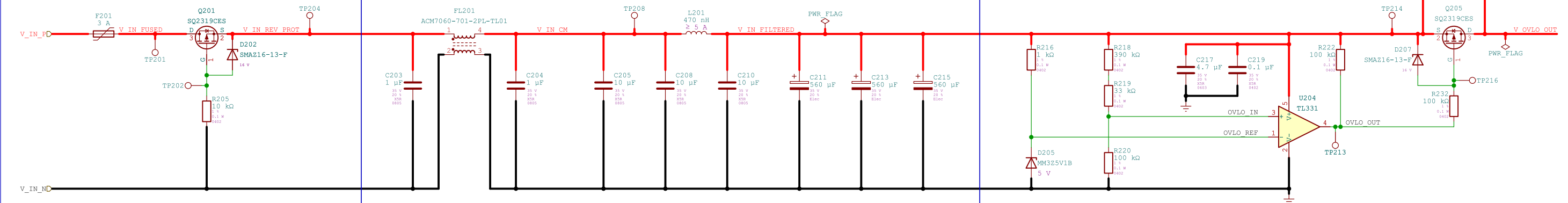
EMI filter

Target cutoff @ 100 kHz with -60 dB @ 2.4 MHz

$$I_{\text{TRANSIENT_MAX}} = 2.5 \text{ A}$$
$$V_{\text{TRANSIENT_MAX}} = 0.05 \text{ V}$$
$$C_{\text{BULK_TOTAL_MIN}} = (1.21 * I_{\text{TRANSIENT_MAX}}^2 * L) / V_{\text{TRANSIENT_MAX}}^2 \sim 1422 \text{ uF}$$

Overvoltage-lockout @ 26.4 V

OVLO_OUT @ 26.4 V $\sim$ 0 V, @ 26.5 V $\sim$ 26.5V

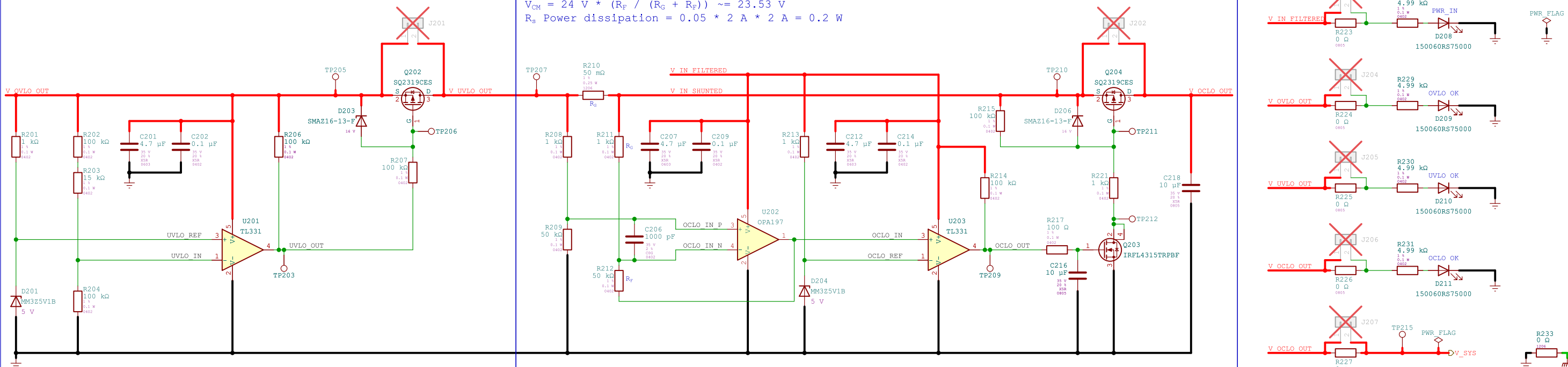


Undervoltage-lockout @ 10.8 V

UVLO_OUT @ 10.8 V $\sim$ 0 V, @ 10.7 V $\sim$ 10.7 V

Overcurrent-lockout @ 2 A

$$\text{OCLO_IN} = I_{\text{IN}} * R_{\text{S}} * (R_{\text{F}} / R_{\text{G}}) \rightarrow @ 0.1 \text{ A} = 0.25 \text{ V}, @ 2 \text{ A} = 5 \text{ V}$$
$$V_{\text{CM}} = 24 \text{ V} * (R_{\text{F}} / (R_{\text{G}} + R_{\text{F}})) \sim 23.53 \text{ V}$$
$$R_{\text{S}} \text{ Power dissipation} = 0.05 * 2 \text{ A} * 2 \text{ A} = 0.2 \text{ W}$$



Notes

- See UVLO.asc for UVLO simulation
- See OVLO.asc for OVLO simulation
- See OCLO.asc for OCLO simulation
- See EMI_FILTER.asc for EMI filter simulation

Sheet: /Power input/

File: power_input.kicad_sch

Title: Linux SBC

Size: A3

Date: 2025-11-07

Rev: v0.0.1

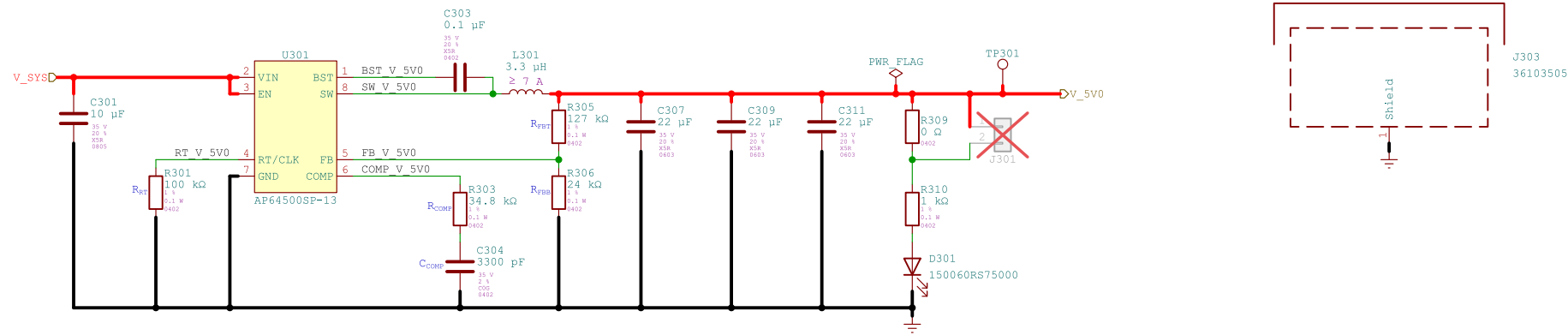
KiCad E.D.A. 9.0.6

Id: 2/13

3. Power generation

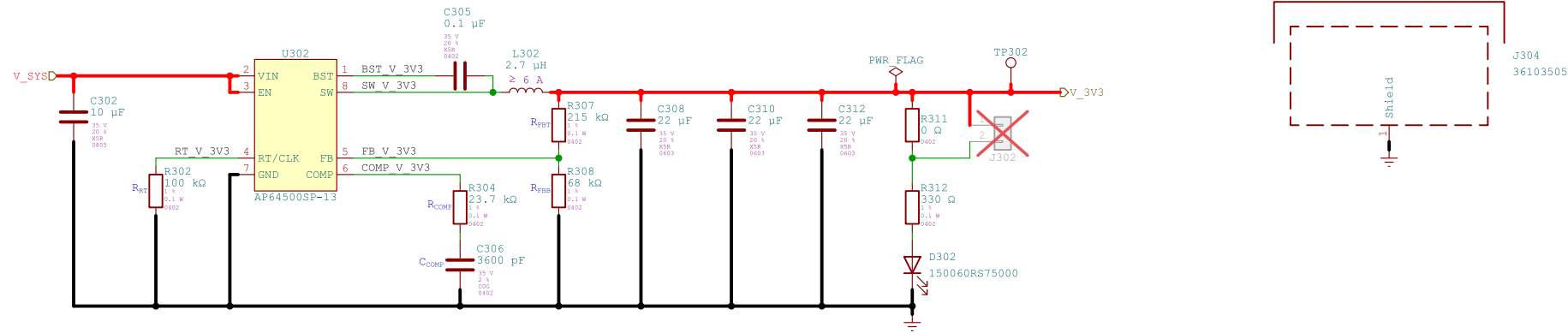
5 V @ 4.5 A

$R_{RT} @ 1000 \text{ kHz} = 100000 / f_{SW[\text{kHz}]} = 100 \text{ k}\Omega$
 $R_{FBT} = R_{FBB} * ((V_{OUT} / 0.8 \text{ V}) - 1) = 24 \text{ k}\Omega * ((5 \text{ V} / 0.8 \text{ V}) - 1) = 126 \text{ k}\Omega$
 $R_{COMP} = 4.67 * 10^3 [\Omega/\text{A}] * f_c * V_{OUT} * C_{OUT} = 4.67 * 10^3 [\Omega/\text{A}] * 15 \text{ kHz} * 5 \text{ V} * 100 \text{ }\mu\text{F} \sim 35 \text{ k}\Omega$
 $C_{COMP} = (V_{OUT} * C_{OUT}) / (I_{OUT} * R_{COMP}) = (5 \text{ V} * 100 \text{ }\mu\text{F}) / (4.5 \text{ A} * 35 \text{ k}\Omega) \sim 3175 \text{ pF}$
 $L = (V_{OUT} * (V_{IN_MAX} - V_{OUT})) / (V_{IN_MAX} * \Delta I_L * f_{SW}) = (5 \text{ V} * (24 \text{ V} - 5 \text{ V})) / (24 \text{ V} * (4.5 \text{ A} * 30 \text{ \%}) * 1000 \text{ kHz}) \sim 2.9 \text{ }\mu\text{H}$
 $I_{L_PEAK} = I_{LOAD} + (\Delta I_L / 2) \sim 5.2 \text{ A}$



3.3 V @ 4 A

$R_{RT} @ 1000 \text{ kHz} = 100000 / f_{SW[\text{kHz}]} = 100 \text{ k}\Omega$
 $R_{FBT} = R_{FBB} * ((V_{OUT} / 0.8 \text{ V}) - 1) = 68 \text{ k}\Omega * ((3.3 \text{ V} / 0.8 \text{ V}) - 1) = 212.5 \text{ k}\Omega$
 $R_{COMP} = 4.67 * 10^3 [\Omega/\text{A}] * f_c * V_{OUT} * C_{OUT} = 4.67 * 10^3 [\Omega/\text{A}] * 15 \text{ kHz} * 3.3 \text{ V} * 100 \text{ }\mu\text{F} \sim 23.1 \text{ k}\Omega$
 $C_{COMP} = (V_{OUT} * C_{OUT}) / (I_{OUT} * R_{COMP}) = (3.3 \text{ V} * 100 \text{ }\mu\text{F}) / (4.5 \text{ A} * 35 \text{ k}\Omega) \sim 3571 \text{ pF}$
 $L = (V_{OUT} * (V_{IN_MAX} - V_{OUT})) / (V_{IN_MAX} * \Delta I_L * f_{SW}) = (3.3 \text{ V} * (24 \text{ V} - 3.3 \text{ V})) / (24 \text{ V} * (4 \text{ A} * 30 \text{ \%}) * 1000 \text{ kHz}) \sim 2.37 \text{ }\mu\text{H}$
 $I_{L_PEAK} = I_{LOAD} + (\Delta I_L / 2) = 4.6 \text{ A}$



Notes

Sheet: /Power generation/
File: power_generation.kicad_sch

Title: Linux SBC

Size: A3 Date: 2025-11-07

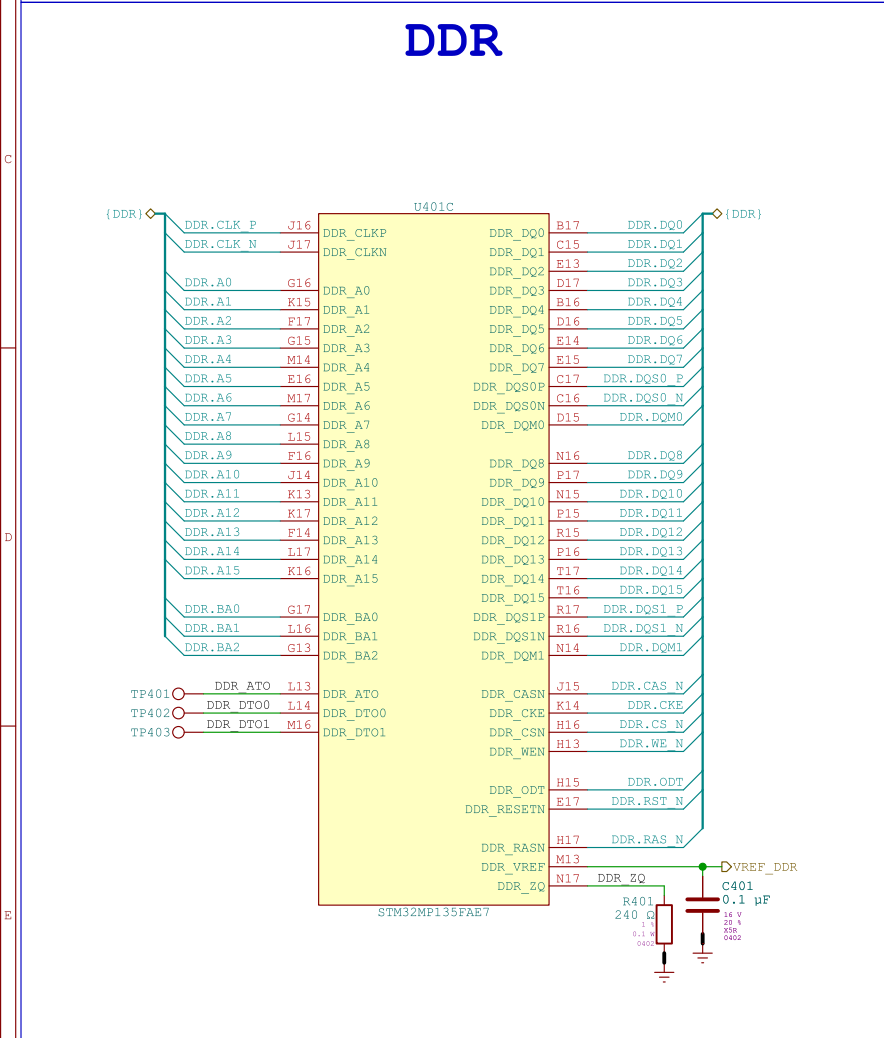
KiCad E.D.A. 9.0.6

Rev: v0.0.1

Id: 3/13

1	2	3	4	5	6	7	8
4 . MPU							

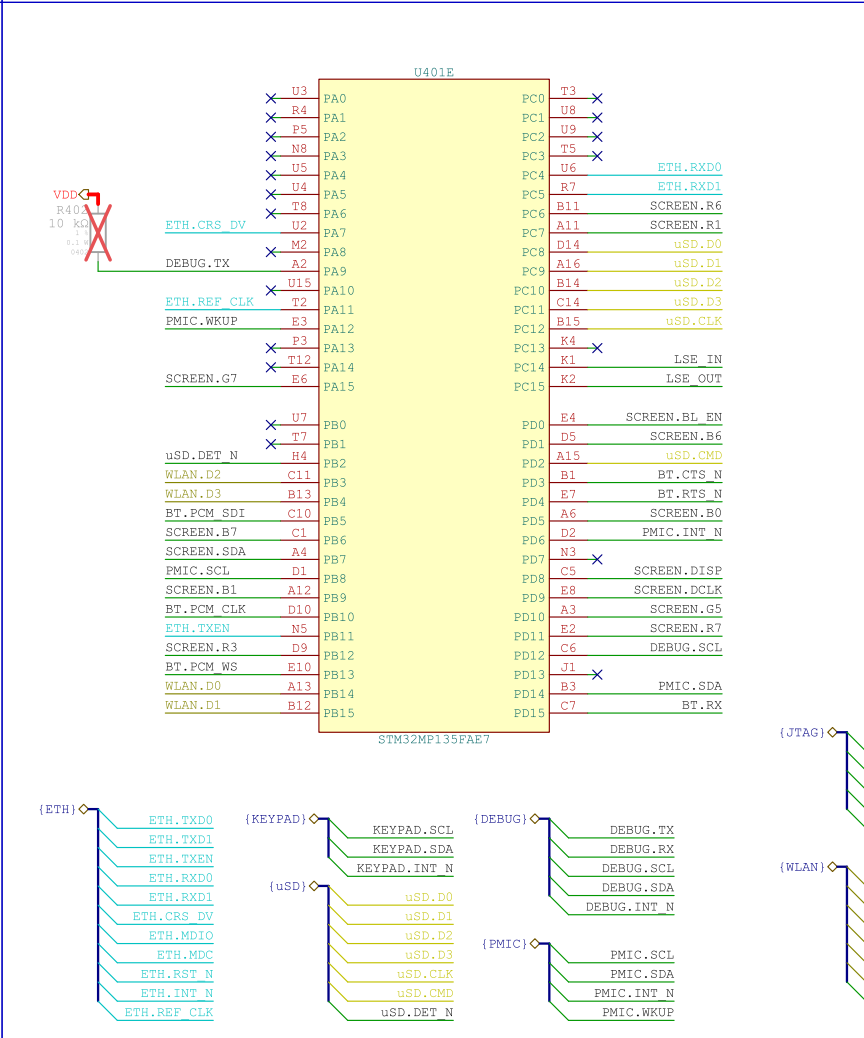
PMIC



Miscellaneous

The diagram illustrates the miscellaneous connections for the STM32MP135FAE7 microcontroller (U401D). Key connections include:

- USB Connections:** USB_DP (T13) and USB_DN (U13) are connected to the USB interface. USB_DP2 (T11) and USB_DM2 (U11) are also shown.
- PMIC Connections:** PMIC_RST_N (U10) is connected to a 0.1 μ F capacitor (C402). USB_RREF (U12) is connected to a 3 k Ω resistor (R403). PMIC_CTRL (P14) is connected to a 10k resistor.
- Power and Control:** PDR_ON, PWR_CPU_ON (R3), PWR_LP (N13), PWR_ON, and PMIC_CTRL (P14) are connected to the power management section.
- Other Pins:** NRST, USB_RREF, and PMIC are also shown.



Boot selection

BOOT2	BOOT1	BOOT0	Boot mode
LOW	LOW	LOW	UART and USB
LOW	LOW	HIGH	Serial NOR-Flash
LOW	HIGH	LOW	eMMC
LOW	HIGH	HIGH	NAND-Flash
HIGH	LOW	LOW	Engineering boot
HIGH	LOW	HIGH	SD card
HIGH	HIGH	LOW	UART and USB
HIGH	HIGH	HIGH	Serial NAND-Flash

VDDC

R405 1 kΩ 5.1 W 402

R406 1 kΩ 5.1 W 402

R407 1 kΩ 5.1 W 402

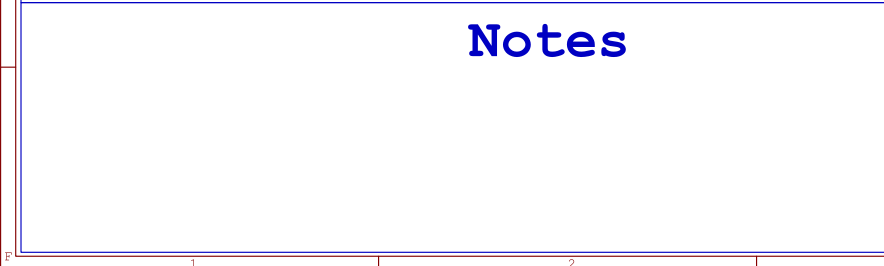
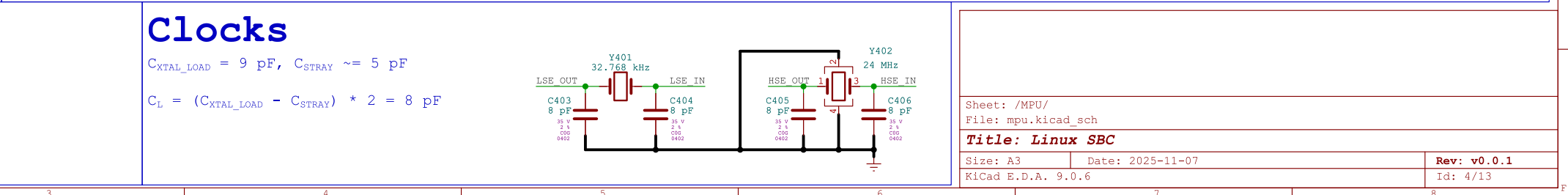
BOOT0

BOOT1

BOOT2

SW401

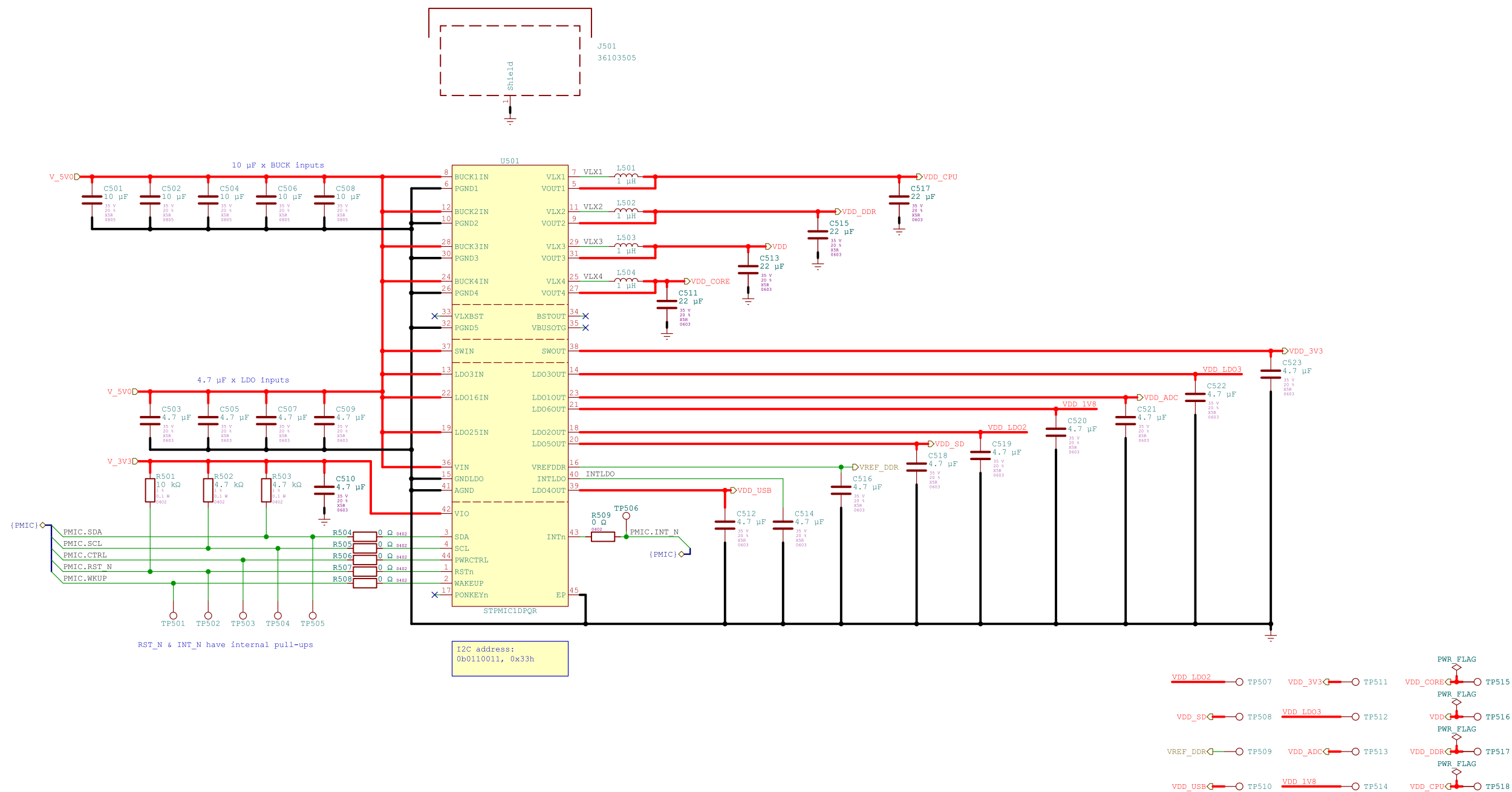
SW_DIP_x03

[illegible]

Clocks

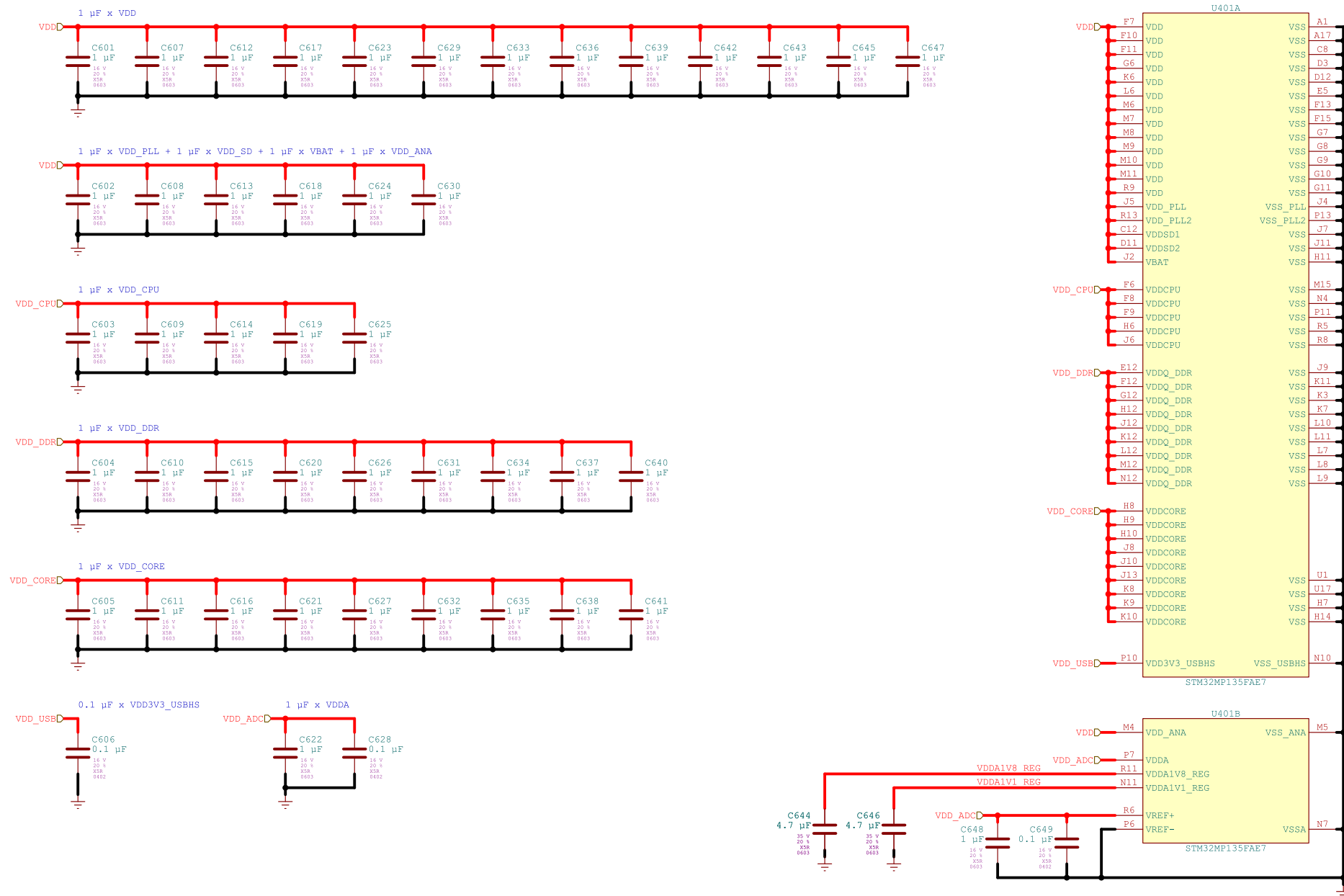
Id: 4/13

4.1. MPU PMIC



Notes

4.2. MPU power



Notes

Sheet: /MPU/MPU power/
File: mpu_power.kicad_sch

Title: Linux SBC

Size: A3 Date: 2025-11-07

Rev: v0.0.1

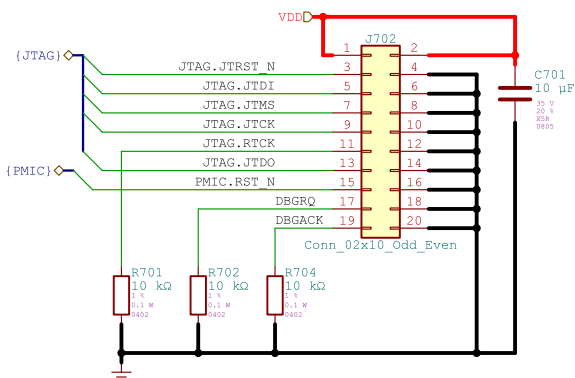
KiCad E.D.A. 9.0.6

Id: 6/13

5. Debug

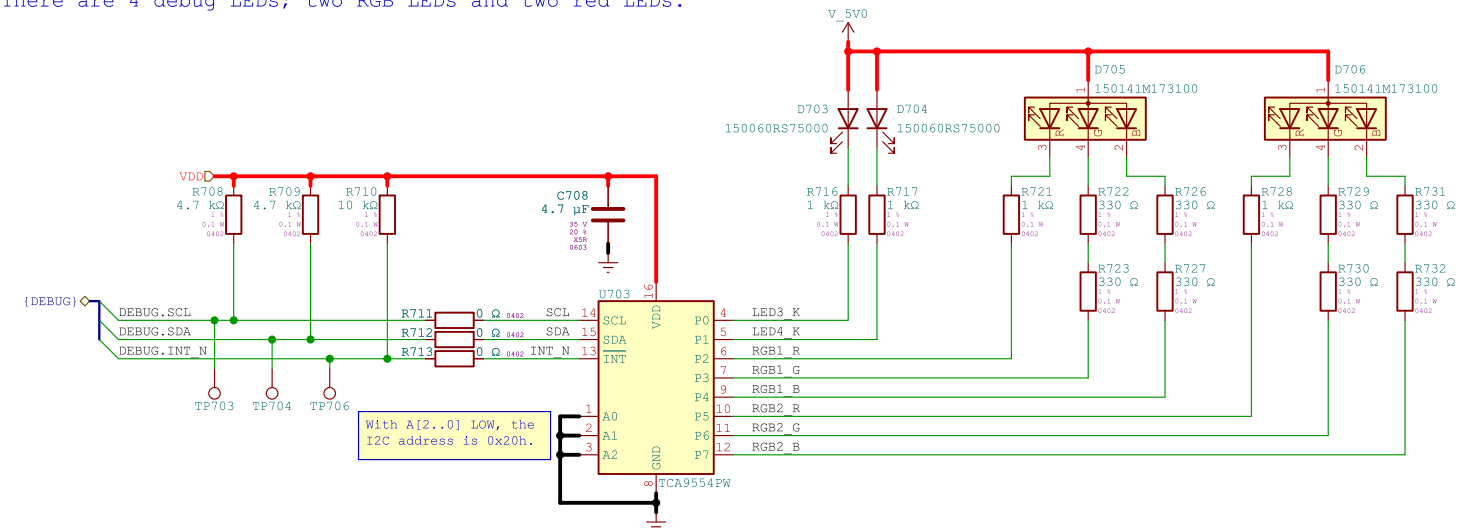
JTAG connector

Connect a JTAG device such as the Segger J-Link to program and debug the device.



LED indicators

There are 4 debug LEDs; two RGB LEDs and two red LEDs.

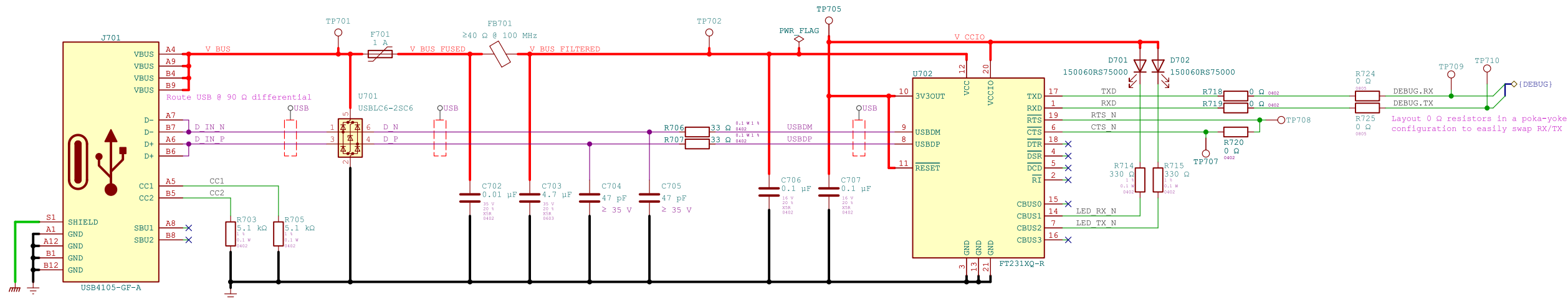


USB to UART bridge

Connect a USB-C cable from a PC to the USB-C receptacle and the debug UART will enumerate as a virtual COM port.

This virtual COM port can be used to program the MPU when the boot selection switch is in the "UART and USB" position.

This virtual COM port can also be used as an external serial interface to the MPU.



Notes

Sheet: /Debug/
File: debug.kicad_sch

Title: Linux SBC

Size: A3 Date: 2025-11-07

KiCad E.D.A. 9.0.6

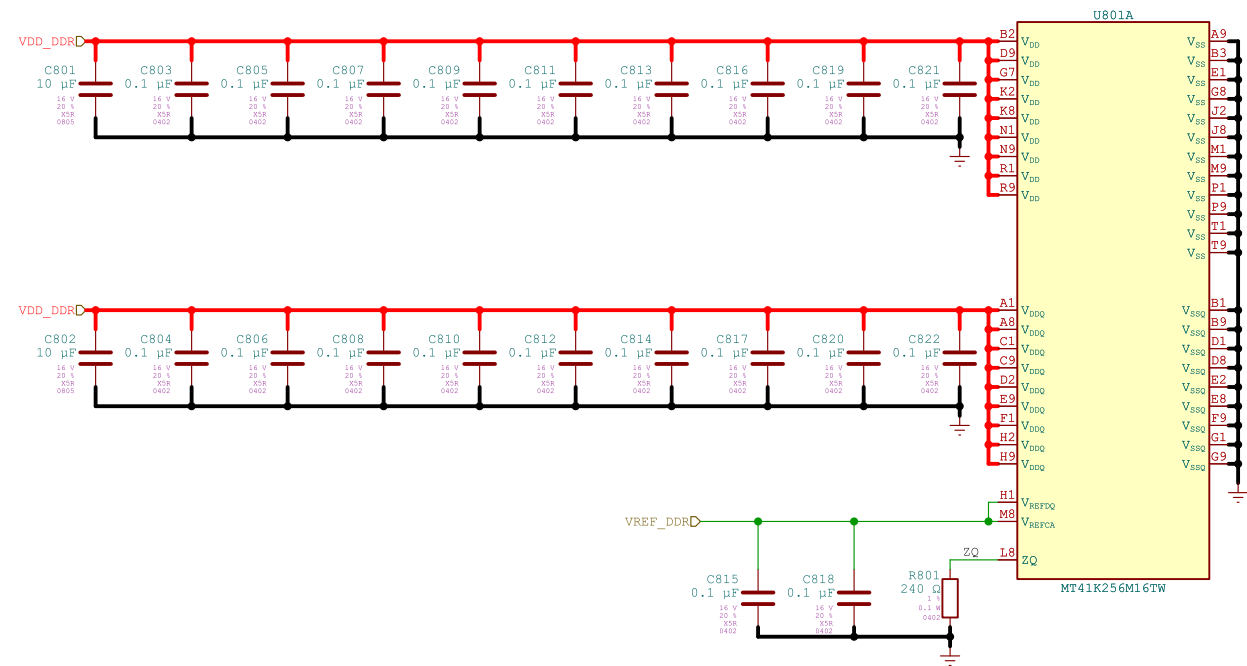
Rev: v0.0.1

Id: 7/13

6. DDR

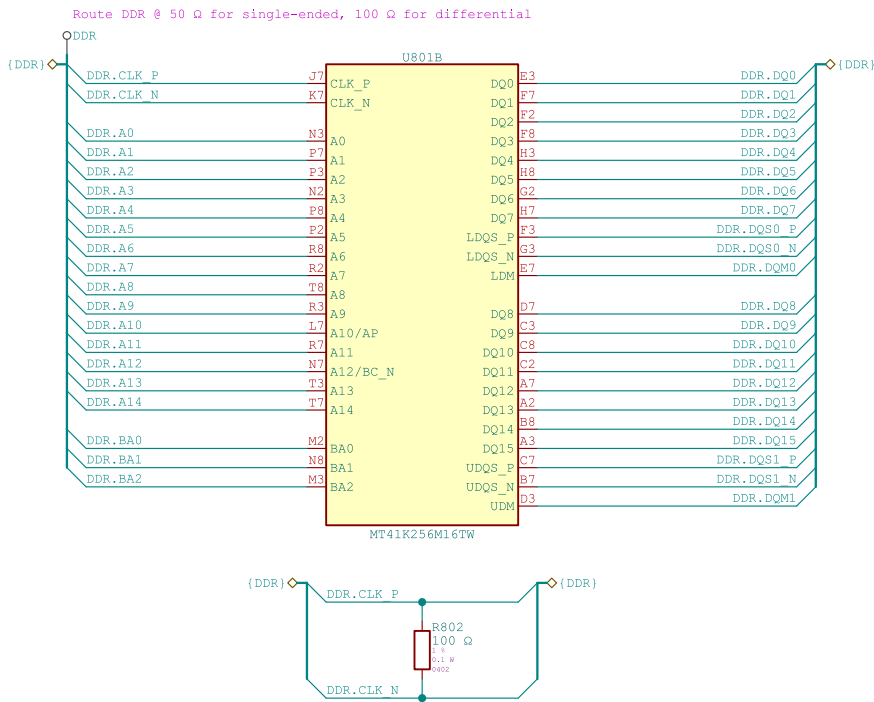
Power supply

10 μ F + 0.1 μ F $\times$ V_{DD} pin
10 μ F + 0.1 μ F $\times$ V_{DDQ} pin
0.1 μ F per V_{REFDQ} and V_{REFCA}



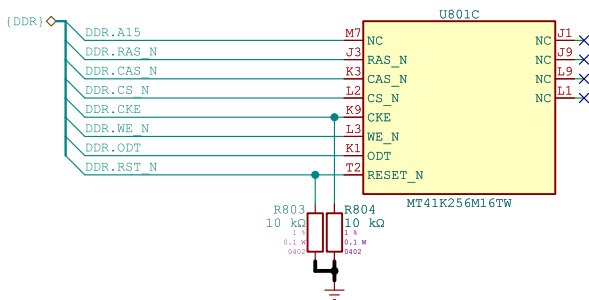
Addressing

Pinout according to STM32MP135F-DK.
On-Die-Termination on DQ/DQS/DM lines.



Control

Pin M7 (NC/A15) used only for 8 Gb DDR3L.



Notes

1	2	3	4	5	6	7	8
7. USB							

USB hub controller

4.7 μ F + 0.1 μ F x VDD pin

Route USB @ 90 Ω differential

Route USB @ 90 Ω differential

USB

USB0

(USB1)

TP901

TP902

TP903

TP904

TP905

CFG_SEL0

CFG_SEL1

CFG_SEL0

CFG_SEL1

NON_REM1

NON_REM0

NON_REM1

NON_REM0

D901

150060RS75000

TABLE 5-1: INITIAL INTERFACE/CONFIGURATION OPTIONS

CFG_SEL[1]	CFG_SEL[0]	Description
0	0	Default configuration: • Strap options enabled • Self-powered operation enabled • Individual power switching • Individual over-current sensing
0	1	The hub is configured externally over SMBus (as an SMBus slave device): • Strap options disabled • All registers configured over SMBus
1	0	Default configuration with the following overrides: • Bus-powered operation
1	1	The hub is configured over 2-wire I ² C EEPROM: • Strap options disabled • All registers configured by I ² C EEPROM

Non-Removable 0 Strap Option: when enabled by package and configuration settings (see Table 5-1 on page 19), this pin will be sampled (in conjunction with NON_REM1[0]) at RESET_N negation to determine if ports 1 and 2 contain permanently attached (non-removable) devices.

Note: When implementing an external LED on this pin, the active state is outlined below and detailed in Section 3.3.1.3, on page 15.

NON_REM1[0] = 00: all ports are removable; LED is active high
NON_REM1[0] = 01: port 1 is non-removable; LED is active low
NON_REM1[0] = 10: ports 1 and 2 are non-removable; LED is active high
NON_REM1[0] = 11: (when available) ports 1, 2, and 3 are non-removable; LED is active low

USB downstream facing ports

Support for up to 4 downstream devices.

Each device is specified to draw up to 0.5 A maximum.

The top diagram illustrates the circuit for Port A[0..1]. It features a power input V_5V0 connected to a 4.7 μF capacitor (C912). The power is regulated by a DC-DC converter (U902, AP22966DC8-7) with feedback capacitors C914 and C916. The output of the converter is connected to the V_BUS_A1 and V_BUS_A2 lines. The signal lines are connected to a USB connector (J901, AU-Y1008-2-BLK) through a USBLC6-2SC6 (U904) and a USBLC6-2SC6 (U906). The connector is also connected to a USB connector (J901, AU-Y1008-2-BLK) through a USBLC6-2SC6 (U906). The connector is also connected to a USB connector (J901, AU-Y1008-2-BLK) through a USBLC6-2SC6 (U906).

The bottom diagram illustrates the circuit for Port B[0..1]. It features a power input V_5V0 connected to a 4.7 μF capacitor (C913). The power is regulated by a DC-DC converter (U903, AP22966DC8-7) with feedback capacitors C915 and C917. The output of the converter is connected to the V_BUS_B1 and V_BUS_B2 lines. The signal lines are connected to a USB connector (J902, AU-Y1008-2-BLK) through a USBLC6-2SC6 (U905) and a USBLC6-2SC6 (U907). The connector is also connected to a USB connector (J902, AU-Y1008-2-BLK) through a USBLC6-2SC6 (U907). The connector is also connected to a USB connector (J902, AU-Y1008-2-BLK) through a USBLC6-2SC6 (U907).

	1	2	3	4	5	6
F						

Sheet: /USB/ File: usb.kicad_sch	
<i>Title: Linux SBC</i>	
Size: A3	Date: 2025-11-07
KiCad E.D.A. 9.0.6	Rev: v0.0.1 Id: 9/13

Title: *Linux SBC*

Size: A3	Date: 2025-11-07	Rev: v0.0.1
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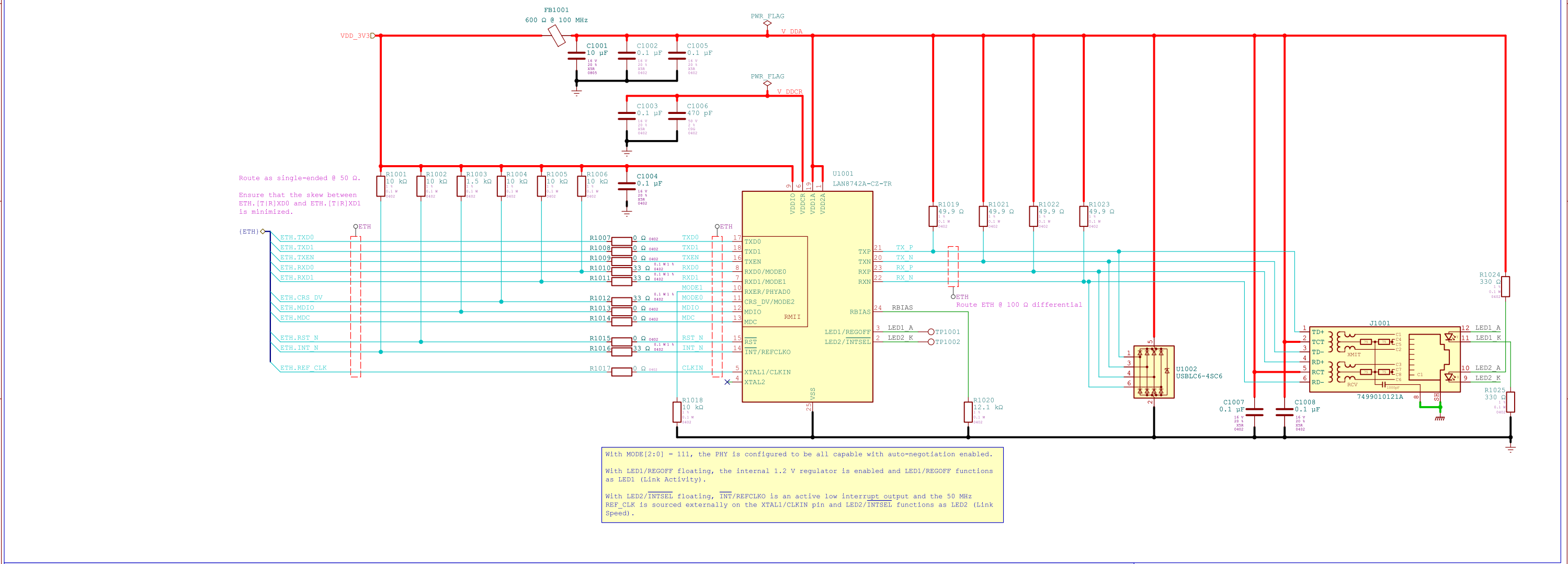
KiCad E.D.A. 9.0.6	Id: 9/13
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A number line from 0 to 10. The segment between 7 and 8 is highlighted in blue.

KiCad E.D.A. 9.0.6	Id: 9/13
--------------------	----------

1	2	3	4	5	6	7	8
8. Ethernet							

- IEEE 802.3u 10/100 Base-T compliant Ethernet.
- Auto-negotiation
- Automatic polarity detection and correction
- Wake-On-Lan support
- Link Activity and Link Speed indicator support
- RMII interface



Notes

Sheet: /Ethernet/ File: ethernet.kicad_sch		
Title: Linux SBC		
Size: A3	Date: 2025-11-07	Rev: v0.0.1
KiCad E.D.A. 9.0.6		Id: 10/13

Size: A3	Date: 2025-11-07	Rev: v0.0.1
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Id: 10/13

3

9. SD card

micro-SD card interface

micro-SD card interface provides the option to provide the operating system image on the SD card and to boot from it.

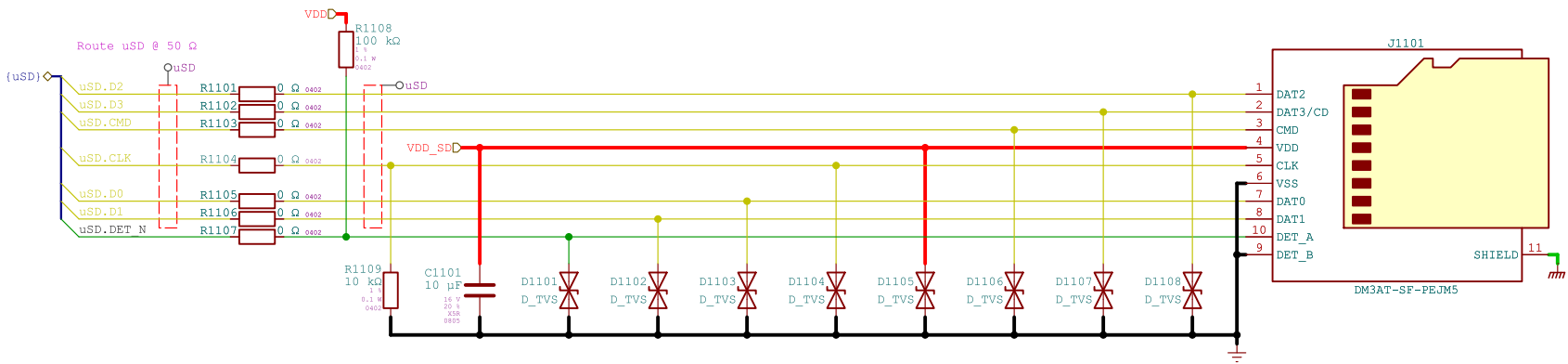
The supported micro-SD card size is up to 1 Terabyte.

The micro-SD card interface is human-interactable and thus has ESD TVS diodes for protection.

The micro-SD card interface is a push-push type and includes insertion detection capability. The insertion detection on the MPU side is active LOW.

When the SD card is not inserted, the DET_A and DET_B pins are disconnected.

When the SD card is connected, the DET_A and DET_B pins are connected.



Notes

Sheet: /SD card/ File: sd_card.kicad_sch		
Title: Linux SBC		
Size: A3	Date: 2025-11-07	Rev: v0.0.1
KiCad E.D.A. 9.0.6		Id: 11/13

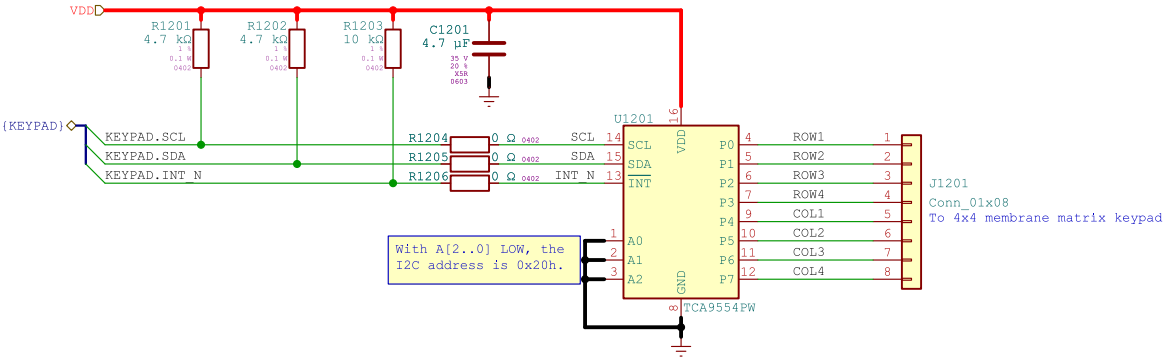
10. User Interface

Keypad interface

They keypad interface is for a 4x4 matrix keypad.

The keypad buttons are polled continuously and button presses are detected by switching rows and columns HIGH/LOW and changing states between INPUT/OUTPUT.

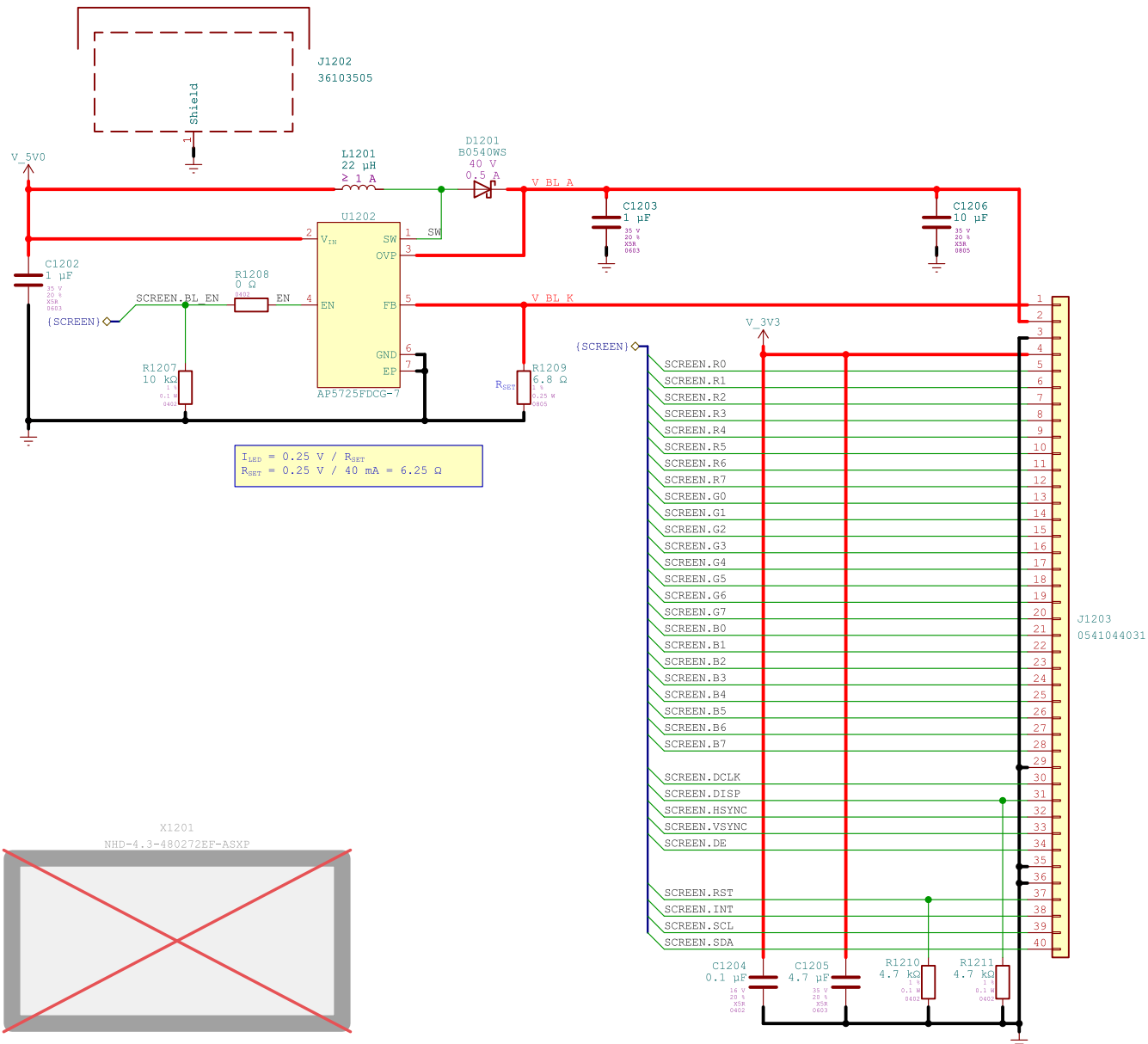
The keypad buttons and their usage is application defined and thus not hard-set.



Graphical interface

The graphical interface is a TFT LCD with a backlight.

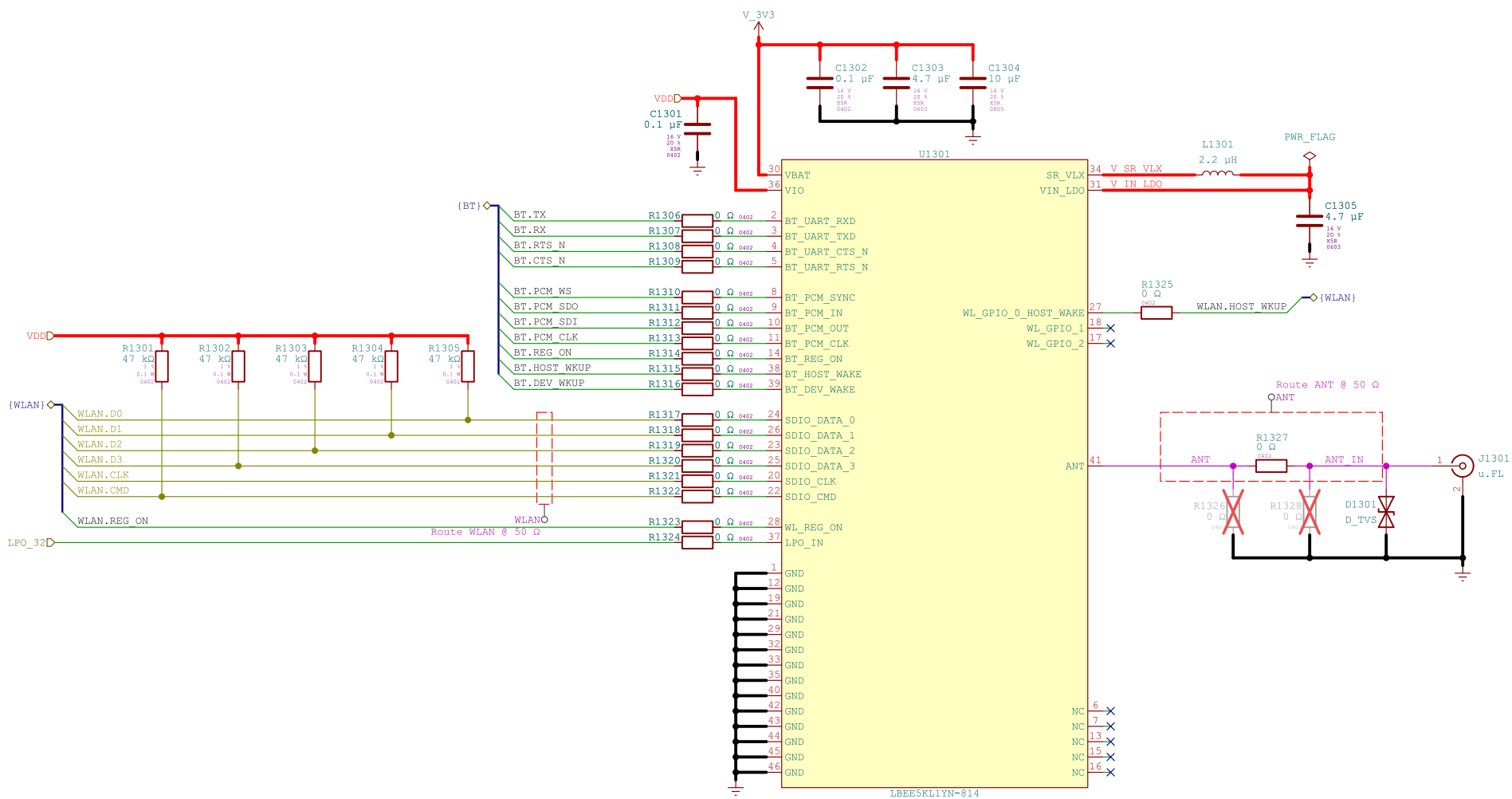
Touchscreen functionality is provided via the I2C interface. The pins can be disconnected if a non-touchscreen TFT LCD is used.



Notes

11. Wireless

WLAN and Bluetooth 5



Notes